David Williams–Young, Ph.D.

Principal Quantum Architect
Microsoft Quantum
Microsoft Corporation
One Microsoft Way, Redmond, WA 98052

PROFESSIONAL INTERESTS

- Architecture and development of software for quantum computing applications.
- Quantum algorithms for chemistry and materials science, including state preparation and circuit synthesis.
- Human-in-the-loop AI systems to accelerate scientific discovery, methods development, and HPC optimization.
- The intersection of high-performance computing and quantum information science.
- Development of high–performance electronic structure methods on emerging architectures.
- Development and microarchitecture optimization of high–performance linear algebra software.
- Leverage of modern C++ to develop clean, reusable, and extensible scientific software.

PROFESSIONAL EXPERIENCE

Principal Quantum Architect ..... September 2026 – Present
Principal Quantum Software Engineer ..... August 2024 – September 2026
Microsoft Quantum
Microsoft Corporation, Redmond, WA

- Lead engineering for Microsoft Quantum's applications software as both engineering manager and technical architect, owning product architecture and direction for QDK/Chemistry and QDK/Analytics.
- Architected QDK/Chemistry, Microsoft's open-source quantum chemistry platform, unifying classical electronic structure, quantum algorithms, and third-party packages in a single modular toolkit.
- Led the integration of AI into the group's R&D workflows across algorithm discovery, methods development, and HPC optimization, compressing development, integration, and optimization work that previously took months into days.
- Established a human-in-the-loop approach to AI-assisted science: built the grounding and steering harness behind these workflows and led the team in applying AI as a force multiplier for deep domain expertise in highly specialized work, not a substitute for it.
- Brought production-grade HPC to quantum workflows, drawing on a decade of GPU and distributed-memory library development (GauXC, MACIS) to deliver QDK/Chemistry's scalable classical simulation layer.
- Previously technical lead for Accelerated DFT, Microsoft's GPU-accelerated cloud service for density functional theory.

Research Scientist (Career) ..... March 2023 – August 2024
Research Scientist (Career-Track) ..... February 2021 – March 2023
Postdoctoral Fellow ..... July 2018 – February 2021
Applied Mathematics and Computational Research Division
Lawrence Berkeley National Laboratory, Berkeley, CA

- Created GauXC, ExchCXX, and MACIS: open-source libraries for GPU-accelerated density functional theory and massively parallel selected configuration interaction, since adopted by NWChemEx, ExaChem, Psi4, MPQC, QDK/Chemistry, and Microsoft's Skala deep-learned exchange-correlation functional.
- Delivered the GPU-accelerated DFT capability of NWChemEx, the DOE Exascale Computing Project's computational chemistry application, targeting DOE leadership computing facilities.
- Authored 27 publications spanning exascale electronic structure, numerical linear algebra, and quantum algorithms; served on the NWChemEx Executive Board.

Graduate Research Assistant ..... July 2013 – July 2018
Graduate Teaching Assistant ..... September 2013 – July 2018
Department of Chemistry
University of Washington, Seattle, WA

EDUCATION

Doctor of Philosophy (Ph.D.), Chemistry ..... May 2018
University of Washington, Seattle, WA
Adviser: Dr. Xiaosong Li
Dissertation: *Towards Efficient and Scalable Electronic Structure Methods for the Treatment of Relativistic Effects and Molecular Response*

Bachelor of Science (B.S., *Magna Cum Laude*), Chemistry, Mathematics ..... May 2013
Indiana University of Pennsylvania, Indiana, PA
Adviser: Dr. Jaeju Ko

COMPUTATIONAL PROFICIENCY

I am a specialist in the following computational areas:
- *Programming Languages:* C++, CUDA
- *Libraries / Paradigms:* OpenMP, MPI, BLAS/(Sca)LAPACK

I am very proficient in the following computational areas:
- *Programming Languages:* C, FORTRAN 77/95/03, HIP, SYCL
- *Scripting Languages:* Python, Julia, Octave, C/Bash Shell
- *Libraries / Paradigms:* TBB, OpenACC
- *Software:* Mathematica, MATLAB

I am the lead architect of the following software packages:
- **QDK/Chemistry**: Open-source toolkit for quantum chemistry applications on quantum computers (Microsoft)
- **QDK/Analytics**: Quantum algorithms for data analytics (Microsoft)

I am the primary developer of the following open-source software packages:
- **GauXC**: Enabling high-performance density functional theory calculations on exascale architectures; the integration engine underlying the Skala deep-learned exchange-correlation functional
- **MACIS**: Massively parallel selected configuration interaction methods
- **ExchCXX**: GPU-accelerated library for the evaluation of exchange-correlation functionals
- **IntegratorXX**: Numerical quadrature for molecular integration grids

I have contributed to the development of the following software packages:
- Accelerated DFT (Microsoft)
- The Chronus Quantum (ChronusQ) Software Package
- NWChemEx
- Psi4
- Massively Parallel Quantum Chemistry (MPQC4)
- TiledArray
- Gaussian

PUBLICATIONS
46 in press publications, 5 preprints
* Indicates equal contribution to published work.
† Indicates that the publication was invited.

Preprints

5. Liu, D.; Zhu, S.; de Jong, W.A.; Fang, D.; **Williams-Young, D.B.**; Yang, C.; "*T-efficient unitary synthesis via block-sparse Hamiltonian simulation*" **2026**. arXiv:2610.18589.
4. Liu, D.; Wang, H.; Zhu, S.; Cong, J.; de Jong, W.A.; **Williams-Young, D.B.**; Yang, C.; "*Exponentially Compressed and Garbage-Free Alias Sampling for Polynomial State Preparation*" **2026**. arXiv:2610.03800.
3. Chen, Y.; Baker, N.A.; Gong, R.; Johnston, C.S.N.; Lackey, B.; Liu, H.; Schmidt, S.; Su, Y.; **Williams-Young, D.B.**; Yang, Y.; "*Clifford-efficient sparse state preparation for molecular wavefunctions*" **2026**. arXiv:2608.20593.
2. Baker, N.A.; Bilodeau, B.; Chen, C.; Chen, Y.; Eckhoff, M.; Efimovskaya, A.; Gasparotto, P.; van Gerwen, P.; Gong, R.; Hoang, K.; Hooshmand, Z.; Jenkins, A.J.; Johnston, C.S.N.; Li, R.R.; Liang, J.; Liu, H.; Mills, A.; Mörchen, M.; Nishibuchi, G.; Sun, C.; Ticehurst, B.; Troyer, M.; Unsleber, J.P.; Wernli, S.; **Williams-Young, D.B.**; Zhang, B.; "*QDK/Chemistry: A Modular Toolkit for Quantum Chemistry Applications*" **2026**. arXiv:2601.15253.
1. Luise, G.; Huang, C.-W.; Vogels, T.; Kooi, D.P.; Ehlert, S.; Lanius, S.; Giesbertz, K.J.H.; Karton, A.; Gunceler, D.; Battaglia, S.; Simm, G.N.C.; Szabó, P.B.; Stanley, M.; Bruinsma, W.P.; Huang, L.; Wei, X.; Garrido Torres, J.; Katbashev, A.; Chavez Zavaleta, R.; Máté, B.; Kaba, S.-O.; Sordillo, R.; Chen, Y.; **Williams-Young, D.B.**; Bishop, C.M.; Hermann, J.; van den Berg, R.; Gori-Giorgi, P.; "*Accurate and scalable exchange-correlation with deep learning*" **2025**. arXiv:2506.14665.

Microsoft Quantum **2024-Present**

1. Ju, F.; Wei, X.; Huang, L.; Jenkins, A.J.; Xia, L.; Zhang, J.; Zhu, J.; Yang, H.; Shao, B.; Dai, P.; **Williams-Young, D.B.**; Mayya, A.; Hooshmand, Z.; Efimovskaya, A.; Baker, N.A.; Troyer, M.; Liu, H.; "*Acceleration without Disruption: DFT Software as a Service*" *J. Chem. Theory Comput.* **2024**, 20(24), 10838–10851.

Lawrence Berkeley National Laboratory **2018-2024**

27. Shen, Y.; Camps, D.; Szasz, A.; Darbha, S.; Klymko, K.; **Williams-Young, D.B.**; Tubman, N.M.; Van Beeumen, R.; "*Estimating Eigenenergies from Quantum Dynamics: A Unified Noise-Resilient Measurement-Driven Approach*" *Quantum* **2025**, 9, 1836.
26. † Blum, V.; Asahi, R.; Autschbach, J.; Bannwarth, C.; Bihlmayer, G.; Blügel, S.; Burns, L.A.; Crawford, T.D.; Dawson, W.; de Jong, W.A.; Draxl, C.; Filippi, C.; Genovese, L.; Giannozzi, P.; Govind, N.; Hammes-Schiffer, S.; Hammond, J.R.; Hourahine, B.; Jain, A.; Kanai, Y.; Kent, P.R.C.; Larsen, A.H.; Lehtola, S.; Li, X.; Lindh, R.; Maeda, S.; Makri, N.; Moussa, J.; Nakajima, T.; Nash, J.A.; Oliveira, M.J.T.; Patel, P.D.; Pizzi, G.; Pourtois, G.; Pritchard, B.P.; Rabani, E.; Reiher, M.; Reining, L.; Ren, X.; Rossi, M.; Schlegel, H.B.; Seriani, N.; Slipchenko, L.V.; Thom, A.; Valeev, E.F.; Van Troeye, B.; Visscher, L.; Vlček, V.; Werner, H.-J.; **Williams-Young, D.B.**; Windus, T.L.; "*Roadmap on methods and software for electronic structure based simulations in chemistry and materials*" *Electron. Struct.* **2024**, 6(4), 042501.
25. Hirsbrunner, M.R.; Mullinax, J.W.; Shen, Y.; **Williams-Young, D.B.**; Klymko, K.; Van Beeumen, R.; Tubman, N.M.; "*A circuit-generated quantum subspace algorithm for the variational quantum eigensolver*" *J. Chem. Phys.* **2024**, 161(16), 164103.

24. Bylaska, E.J.; Panyala, A.; Bauman, N.P.; Peng, B.; Pathak, H.; Mejia-Rodriguez, D.; Govind, N.; **Williams-Young, D.B.**; Aprà, E.; Bagusetty, A.; Mutlu, E.; Jackson, K.A.; Baruah, T.; Yamamoto, Y.; Pederson, M.R.; Withanage, K.P.K.; Pedroza-Montero, J.N.; Bilbrey, J.A.; Choudhury, S.; Firoz, J.; Herman, K.M.; Xantheas, S.S.; Rigor, P.; Vila, F.D.; Rehr, J.J.; Fung, M.; Grofe, A.; Johnston, C.; Baker, N.; Kaneko, K.; Liu, H.; Kowalski, K.; "*Electronic structure simulations in the cloud computing environment*" *J. Chem. Phys.* **2024**, 161(15), 150902.

23. Kovtun, M.; Lambros, E.; Liu, A.; Tang, D.; **Williams-Young, D.B.**; Li, X.; "*Accelerating Relativistic Exact-Two-Component Density Functional Theory Calculations with Graphical Processing Units*" *J. Chem. Theory Comput.* **2024**, 20(18), 7694–7699.

22. Poole, D.; **Williams-Young, D.B.**; Jiang, A.; Glick, Z.L.; Sherrill, C.D.; "*A modular, composite framework for the utilization of reduced-scaling Coulomb and exchange construction algorithms: Design and implementation*" *J. Chem. Phys.* **2024**, 161(5), 052503.

21. † Burton, H.G.A.; Dong, S.S.; Ghosh, S.; Gu, B.; Jackson, N.E.; Keefer, D.; Lu, Y.; Monroe, J.I.; Peng, B.; Pieri, E.; Spackman, P.R.; Vacher, M.; Vuckovic, S.; **Williams-Young, D.B.**; Yang, Z.J.; Yue, S.; Zerze, G.H.; Zhu, T.; "*JCTC Early Career Board Selects*" *J. Chem. Theory Comput.* **2024**, 20(14), 5785–5787.

20. Alvertis, A.M.; **Williams-Young, D.B.**; Bruneval, F.; Neaton, J.B.; "*Influence of Electronic Correlations on Electron–Phonon Interactions of Molecular Systems with the GW and Coupled Cluster Methods*" *J. Chem. Theory Comput.* **2024**, 20(14), 6175–6183.

19. **Williams-Young, D.B.**; Yuwono, S.H.; DePrince III, A.E.; Yang, C.; "*Approximate Exponential Integrators for Time-Dependent Equation-of-Motion Coupled Cluster Theory*" *J. Chem. Theory Comput.* **2023**, 19(24), 9177–9186.

18. Di Felice, R.; Mayes, M.L.; Richard, R.M.; **Williams-Young, D.B.**; Chan, G.K.-L.; de Jong, W.A.; Govind, N.; Head-Gordon, M.; Hermes, M.R.; Kowalski, K.; Li, X.; Lischka, H.; Mueller, K.T.; Mutlu, E.; Niklasson, A.M.N.; Pederson, M.R.; Peng, B.; Shepard, R.; Valeev, E.F.; van Schilfgaarde, M.; Vlaisavljevich, B.; Windus, T.L.; Xantheas, S.S.; Zhang, X.; Zimmerman, P.M.; "*A Perspective on Sustainable Computational Chemistry Software Development and Integration*" *J. Chem. Theory Comput.* **2023**, 19(20), 7056–7076.

17. Ko, T.; Heindel, J.P.; Guan, X.; Head-Gordon, T.; **Williams-Young, D.B.**; Yang, C.; "*Using Diffusion Maps to Analyze Reaction Dynamics for a Hydrogen Combustion Benchmark Dataset*" *J. Chem. Theory Comput.* **2023**, 19(17), 5872–5885.

16. Shen, Y.; Klymko, K.; Sud, J.; **Williams-Young, D.B.**; de Jong, W.A.; Tubman, N.M.; "*Real-Time Krylov Theory for Quantum Computing Algorithms*" *Quantum* **2023**, 7, 1066.

15. † **Williams-Young, D.B.**; Asadchev, A.; Popovici, D.T.; Clark, D.; Waldrop, J.; Windus, T.L.; Valeev, E.F.; de Jong, W.A.; "*Distributed memory, GPU accelerated Fock construction for hybrid, Gaussian basis density functional theory*" *J. Chem. Phys.* **2023**, 158(23), 234104.

14. † **Williams-Young, D.B.**; Tubman, N.M.; Mejuto-Zaera, C.; de Jong, W.A.; "*A parallel, distributed memory implementation of the adaptive sampling configuration interaction method*" *J. Chem. Phys.* **2023**, 158(21), 214109.

13. Gomes, N.; **Williams-Young, D.B.**; de Jong, W.A.; "*Computing the Many-Body Green's Function with Adaptive Variational Quantum Dynamics*" *J. Chem. Theory Comput.* **2023**, 19(11), 3313–3323.

12. Richard, R.M.; Keipert, K.; Waldrop, J.; Keçeli, M.; **Williams-Young, D.B.**; Bair, R.; Boschen, J.; Crandall, Z.; Gasperich, K.; Mahmud, Q.I.; Panyala, A.; Valeev, E.; van Dam, H.; de Jong, W.A.; Windus, T.L.; "*PluginPlay: Enabling exascale scientific software one module at a time*" *J. Chem. Phys.* **2023**, 158(18), 184801.

11. Sid-Lakhdar, W.; Cayrols, S.; Bielich, D.; Abdelfattah, A.; Luszczek, P.; Gates, M.; Tomov, S.; Johansen, H.; **Williams-Young, D.B.**; Davis, T.; Dongarra, J.; Anzt, H.; "*PAQR: Pivoting Avoiding QR factorization*" In *2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, **2023**, pp. 322–332.

10. Mejuto-Zaera, C.; Tzeli, D.; **Williams-Young, D.B.**; Tubman, N.M.; Matoušek, M.; Brabec, J.; Veis, L.; Xantheas, S.S.; de Jong, W.A.; "*The Effect of Geometry, Spin, and Orbital Optimization in Achieving Accurate, Correlated Results for Iron–Sulfur Cubanes*" *J. Chem. Theory Comput.* **2022**, 18(2), 687–702.
9. **Williams-Young, D.B.**; Bagusetty, A.; de Jong, W.A.; Doerfler, D.; van Dam, H.J.J.; Vázquez-Mayagoitia, Á.; Windus, T.L.; Yang, C.; "*Achieving performance portability in Gaussian basis set density functional theory on accelerator based architectures in NWChemEx*" *Parallel Comput.* **2021**, 108, 102829.
8. Bez, J.L.; Tang, H.; Xie, B.; **Williams-Young, D.B.**; Latham, R.; Ross, R.; Oral, S.; Byna, S.; "*I/O Bottleneck Detection and Tuning: Connecting the Dots using Interactive Log Analysis*" In *2021 IEEE/ACM Sixth International Parallel Data Systems Workshop (PDSW)*, **2021**, pp. 15–22.
7. Ahmed, H.; **Williams-Young, D.B.**; Ibrahim, K.Z.; Yang, C.; "*Performance Modeling and Tuning for DFT Calculations on Heterogeneous Architectures*" In *2021 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, **2021**, pp. 714–722.
6. Kowalski, K.; Bair, R.; Bauman, N.P.; Boschen, J.S.; Bylaska, E.J.; Daily, J.; de Jong, W.A.; Dunning, T.; Govind, N.; Harrison, R.J.; Keçeli, M.; Keipert, K.; Krishnamoorthy, S.; Kumar, S.; Mutlu, E.; Palmer, B.; Panyala, A.; Peng, B.; Richard, R.M.; Straatsma, T.P.; Sushko, P.; Valeev, E.F.; Valiev, M.; van Dam, H.J.J.; Waldrop, J.M.; **Williams-Young, D.B.**; Yang, C.; Zalewski, M.; Windus, T.L.; "*From NWChem to NWChemEx: Evolving with the Computational Chemistry Landscape*" *Chem. Rev.* **2021**, 121(8), 4962–4998.
5. † **Williams-Young, D.B.**; de Jong, W.A.; van Dam, H.J.J.; Yang, C.; "*On the Efficient Evaluation of the Exchange Correlation Potential on Graphics Processing Unit Clusters*" *Front. Chem.* **2020**, 8, 581058.
4. Yang, C.; Brabec, J.; Veis, L.; **Williams-Young, D.B.**; Kowalski, K.; "*Solving Coupled Cluster Equations by the Newton Krylov Method*" *Front. Chem.* **2020**, 8, 590184.
3. **Williams-Young, D.B.**; Beckman, P.G.; Yang, C.; "*A Shift Selection Strategy for Parallel Shift-invert Spectrum Slicing in Symmetric Self-consistent Eigenvalue Computation*" *ACM Trans. Math. Softw.* **2020**, 46(4), 35:1–35:31.
2. **Williams-Young, D.B.**; Yang, C.; "*Parallel Shift-Invert Spectrum Slicing on Distributed Architectures with GPU Accelerators*" In *49th International Conference on Parallel Processing - ICPP*, **2020**, pp. 1–11.
1. Peng, B.; Van Beeumen, R.; **Williams-Young, D.B.**; Kowalski, K.; Yang, C.; "*Approximate Green's Function Coupled Cluster Method Employing Effective Dimension Reduction*" *J. Chem. Theory Comput.* **2019**, 15(5), 3185–3196.

University of Washington 2013-2018

18. † **Williams-Young, D.B.**; Petrone, A.; Sun, S.; Stetina, T.F.; Lestrange, P.; Hoyer, C.E.; Nascimento, D.R.; Koulias, L.; Wildman, A.; Kasper, J.; Goings, J.J.; Ding, F.; DePrince, A.E.; Valeev, E.F.; Li, X.; "*The Chronus Quantum software package*" *WIREs Comput. Mol. Sci.* **2020**, 10(2), e1436.
17. Sun, S.; **Williams-Young, D.B.**; Stetina, T.F.; Li, X.; "*Generalized Hartree–Fock with Nonperturbative Treatment of Strong Magnetic Fields: Application to Molecular Spin Phase Transitions*" *J. Chem. Theory Comput.* **2019**, 15(1), 348–356.
16. Koulias, L.N.; **Williams-Young, D.B.**; Nascimento, D.R.; DePrince, A.E.; Li, X.; "*Relativistic Real-Time Time-Dependent Equation-of-Motion Coupled-Cluster*" *J. Chem. Theory Comput.* **2019**, 15(12), 6617–6624.
15. Sun, S.; Beck, R.A.; **Williams-Young, D.B.**; Li, X.; "*Simulating Magnetic Circular Dichroism Spectra with Real-Time Time-Dependent Density Functional Theory in Gauge Including Atomic Orbitals*" *J. Chem. Theory Comput.* **2019**, 15(12), 6824–6831.

14. Stetina, T.F.; Sun, S.; **Williams-Young, D.B.**; Li, X.; "*Modeling Magneto-Photoabsorption Using Time-Dependent Complex Generalized Hartree-Fock*" *ChemPhotoChem* **2019**, 3(9), 739–746.
13. Hoyer, C.E.; **Williams-Young, D.B.**; Huang, C.; Li, X.; "*Embedding non-collinear two-component electronic structure in a collinear quantum environment*" *J. Chem. Phys.* **2019**, 150(17), 174114.
12. Sun, S.; **Williams-Young, D.B.**; Li, X.; "*An ab Initio Linear Response Method for Computing Magnetic Circular Dichroism Spectra with Nonperturbative Treatment of Magnetic Field*" *J. Chem. Theory Comput.* **2019**, 15(5), 3162–3169.
11. Petrone, A.*; **Williams-Young, D.B.***; Sun, S.; Stetina, T.F.; Li, X.; "*An efficient implementation of two-component relativistic density functional theory with torque-free auxiliary variables*" *Eur. Phys. J. B* **2018**, 91(7), 169.
10. Kasper, J.M.; **Williams-Young, D.B.**; Vecharynski, E.; Yang, C.; Li, X.; "*A Well-Tempered Hybrid Method for Solving Challenging Time-Dependent Density Functional Theory (TDDFT) Systems*" *J. Chem. Theory Comput.* **2018**, 14(4), 2034–2041.
9. Lestrange, P.J.; **Williams-Young, D.B.**; Petrone, A.; Jiménez-Hoyos, C.A.; Li, X.; "*Efficient Implementation of Variation after Projection Generalized Hartree–Fock*" *J. Chem. Theory Comput.* **2018**, 14(2), 588–596.
8. Van Beeumen, R.; **Williams-Young, D.B.**; Kasper, J.M.; Yang, C.; Ng, E.G.; Li, X.; "*Model Order Reduction Algorithm for Estimating the Absorption Spectrum*" *J. Chem. Theory Comput.* **2017**, 13(10), 4950–4961.
7. Barclay, M.S.; Quincy, T.J.; **Williams-Young, D.B.**; Caricato, M.; Elles, C.G.; "*Accurate Assignments of Excited-State Resonance Raman Spectra: A Benchmark Study Combining Experiment and Theory*" *J. Phys. Chem. A* **2017**, 121(41), 7937–7946.
6. Egidi, F.*; **Williams-Young, D.B.***; Baiardi, A.*; Bloino, J.; Scalmani, G.; Frisch, M.J.; Li, X.; Barone, V.; "*Effective Inclusion of Mechanical and Electrical Anharmonicity in Excited Electronic States: VPT2-TDDFT Route*" *J. Chem. Theory Comput.* **2017**, 13(6), 2789–2803.
5. Petrone, A.*; **Williams-Young, D.B.***; Lingerfelt, D.B.; Li, X.; "*Ab Initio Excited-State Transient Raman Analysis*" *J. Phys. Chem. A* **2017**, 121(20), 3958–3965.
4. Petrone, A.; Lingerfelt, D.B.; **Williams-Young, D.B.**; Li, X.; "*Ab Initio Transient Vibrational Spectral Analysis*" *J. Phys. Chem. Lett.* **2016**, 7(22), 4501–4508.
3. **Williams-Young, D.B.**; Goings, J.J.; Li, X.; "*Accelerating Real-Time Time-Dependent Density Functional Theory with a Nonrecursive Chebyshev Expansion of the Quantum Propagator*" *J. Chem. Theory Comput.* **2016**, 12(11), 5333–5338.
2. **Williams-Young, D.B.**; Egidi, F.; Li, X.; "*Relativistic Two-Component Particle–Particle Tamm–Dancoff Approximation*" *J. Chem. Theory Comput.* **2016**, 12(11), 5379–5384.
1. Lingerfelt, D.B.; **Williams-Young, D.B.**; Petrone, A.; Li, X.; "*Direct ab Initio (Meta-)Surface-Hopping Dynamics*" *J. Chem. Theory Comput.* **2016**, 12(3), 935–945.

HONORS

CCG Excellence Award for Graduate Students — The Chemical Computing Group (2017)
MolSSI Software Fellow — Molecular Sciences Software Institute (2017-2018)
Lloyd E. and Florence M. West Fellowship in Chemistry — Lloyd E. and Florence M. West (2016)
Excellence in Chemistry Graduate Fellowship Award (ECGFA) — University of Washington (2013)

PROFESSIONAL SERVICE

Leadership Activities

Journal of Chemical Theory and Computation Early Career Board (2024)
NWChemEx Executive Board (2023-2024)

Journal Review

Molecular Physics, *The Journal of Chemical Physics*, *The International Journal of Quantum Chemistry*, *Computer Physics Communications*, *The Journal of Chemical Theory and Computation*, *Journal of Computational Physics*, *The Journal of Physical Chemistry*, *The Journal of Physical Chemistry A*

Proposal Review

NSF Office of Advanced Cyberinfrastructure, *DOE Office of Science, Office of Basic Energy Sciences*, *Natural Sciences and Engineering Research Council of Canada*

Conference Symposium Organization

SIAM Conference on Computational Science and Engineering, *SIAM Conference on Parallel Processing for Scientific Computing*

MEMBERSHIPS

Alpha Chi Sigma (AXΣ), ΓT Chapter, Professional Chemistry Fraternity
American Chemical Society (ACS), Computers in Chemistry Division
American Physical Society (APS)
Society for Industrial and Applied Mathematics (SIAM), Activity Group on Computational Science and Engineering